

Multi-Channel Scheduling Algorithms for Fast Aggregated Convergecast in Sensor Networks

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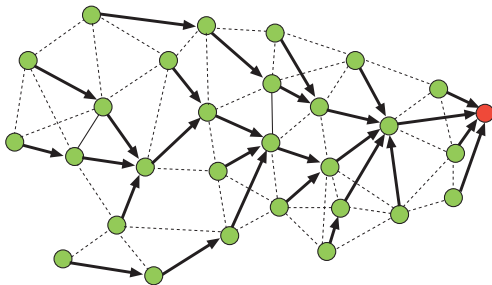
Outline

- Introduction
- Problem Formulation
- Solutions
- Conclusions

Motivation

Aggregated Data Collection:

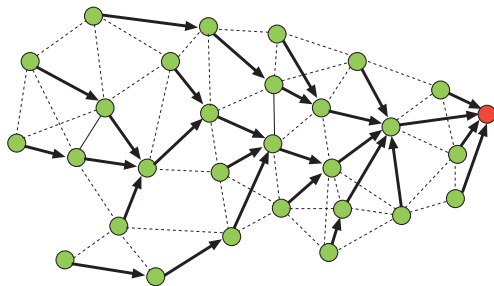
- An **arbitrary deployment** of nodes that **periodically** generate data
- Critical continuous-monitoring applications (e.g., surveillance, environmental events: wildfire, oil/gas leak, permafrost, etc.)



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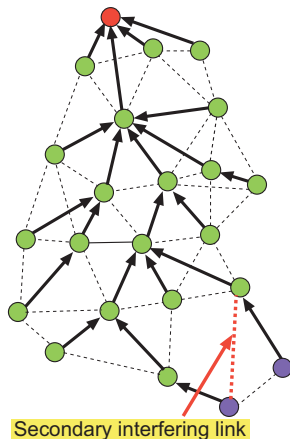
Goal:

What is the **fastest rate** at which **aggregated** data can be streamed from the sources to the sink (using **multiple frequency channels**)?

Model and Assumptions

Model and Assumptions

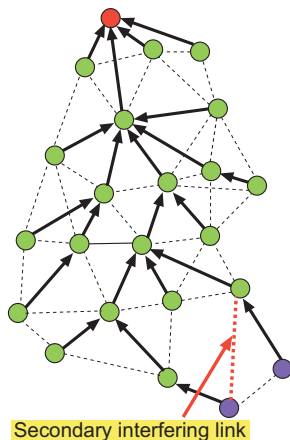
■ Graph-based interference model



e.g., monitoring applications with fast & timely delivery requirements.

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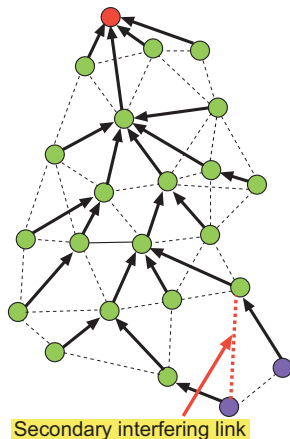
- Graph-based interference model
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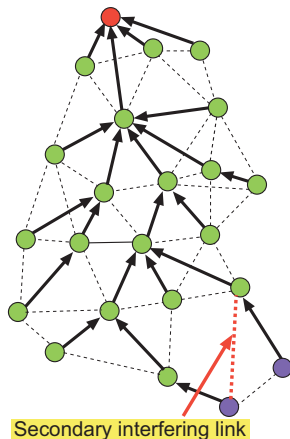
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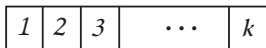
- Graph-based interference model
- Half-duplex, single transceiver
- Receiver-based Frequency Assignment
- Non-interfering orthogonal channels



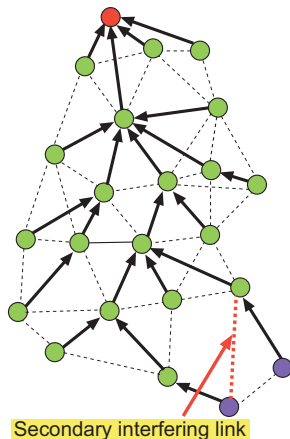
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Model and Assumptions

- Graph-based interference model
- Half-duplex, single transceiver
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- Non-interfering orthogonal channels
- TDMA, periodic scheduling
(- every node generates a single packet at the beginning of each frame
- every node transmits during only a single slot)



Schedule length



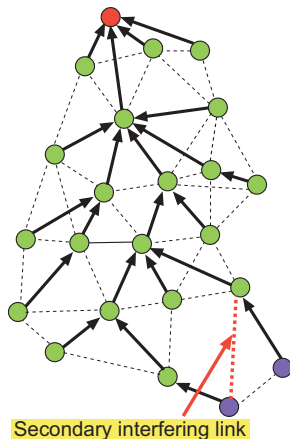
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- Non-interfering orthogonal channels
- TDMA, periodic scheduling
(- every node generates a single packet at the beginning of each frame
- every node transmits during only a single slot)
- In-network perfect aggregation
(*distributive, algebraic*
- aggregation happens at each hop)



Schedule length



e.g., monitoring applications with fast & timely delivery requirements.

Convergecast and Benefits of Multiple Frequencies

Convergecast:

A **many-to-one** communication pattern: from all the nodes to a sink (opposite of broadcast).

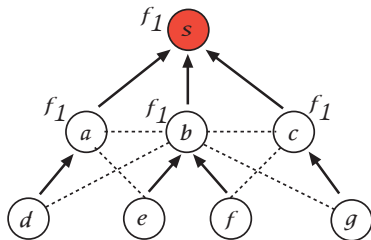


Figure: Single frequency.

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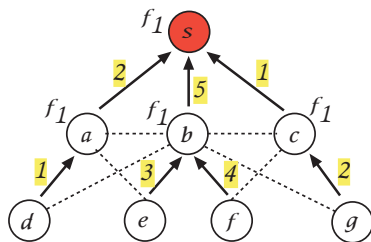


Figure: Schedule length is 5.

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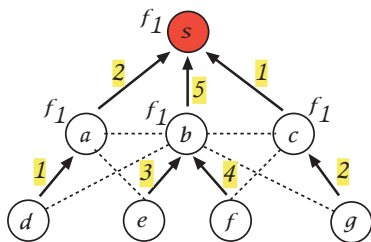


Figure: Schedule length is 5.

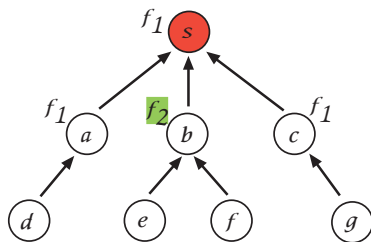


Figure: Two frequencies.

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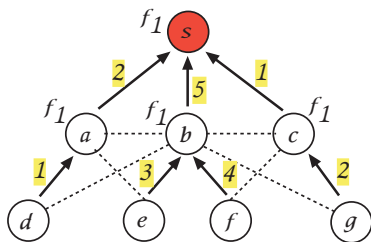


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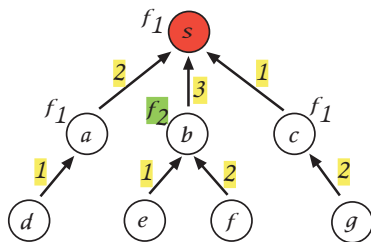


Figure: Schedule length is 3.

Convergecast and Benefits of Multiple Frequencies

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	Frame 1			Frame 2		
Receiver	Slot 1	Slot 2	Slot 3	Slot 1	Slot 2	Slot 3
<i>s</i>						
<i>a</i>						
<i>b</i>						
<i>c</i>						

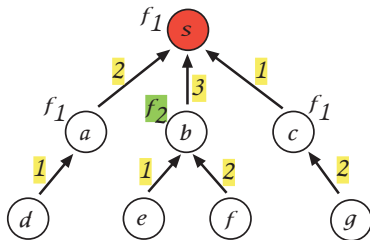


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a	<i>d</i>			<i>d</i>		
b	<i>e</i>	<i>f</i>		<i>e</i>	<i>f</i>	
c		<i>g</i>			<i>g</i>	

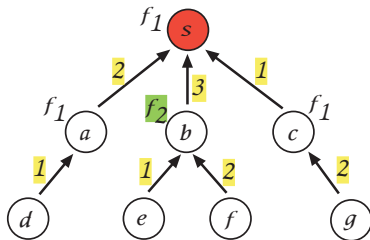
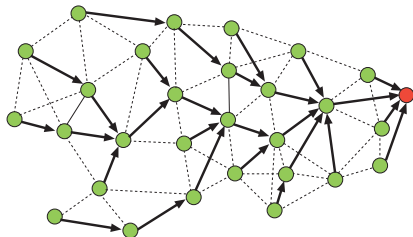


Figure: Schedule length is 3.

Problem Formulation

Given:



A set of nodes, arbitrarily deployed (possibly even *worst case*), a set of frequencies, and a routing (spanning) tree rooted at the sink.

Define:

Aggregated Throughput =
 $1/\text{Schedule Length}$

Joint Frequency and Time Slot Assignment:

Assign a *frequency* to each receiver and a *time slot* to each edge in the routing tree, such that the *aggregated throughput* at the sink is *maximized*.

Scheduling Complexity

Question:

Given a spanning tree T on an arbitrary graph G , is there an assignment of time slots using *at most q frequencies* such that the *schedule length is at most p* ?

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Theorem

Joint Frequency and Time Slot Scheduling is NP-complete on general graphs (and possibly even on disk graphs).

Proof: Reduction from *Vertex Color*.

Approach

Split the *Joint Frequency and Time Slot Assignment* into two subproblems:

- Frequency Assignment
- Time Slot Assignment

Two Scenarios

- With **unlimited** frequencies
- With a **limited** number of frequencies

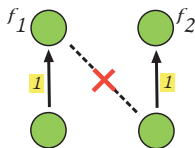
Contributions: Approximation and Randomized algorithms with provable worst-case bounds.

Scenario One: Unlimited Frequencies

Frequency Assignment: Complexity

Question:

Given a spanning tree T on an *arbitrary graph* $G = (V, E)$, can we find the *minimum number of frequencies* that can be assigned to the receivers of T such that *all* the secondary interfering links are removed?
(*Minimum Frequency Assignment*)

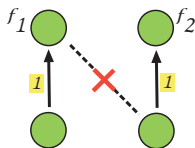


Secondary interfering link

Frequency Assignment: Complexity

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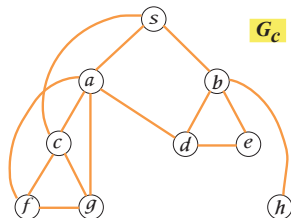
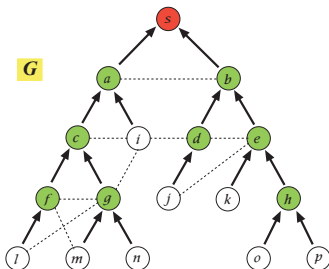
Minimum Frequency Assignment is NP-hard on general graphs.

Proof: Reduction from *Vertex Color*.

Frequency Assignment: Upper Bound

Constraint Graph:

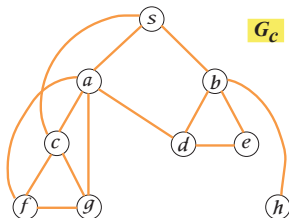
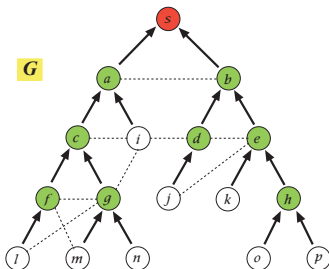
- For each receiver in G create a node in G_c .
- Create an edge between two nodes in G_c if the corresponding receivers in G form secondary interfering links.



Frequency Assignment: Upper Bound

Constraint Graph:

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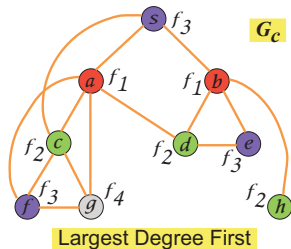
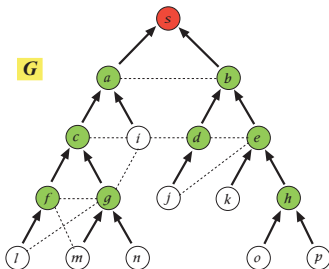
Lemma:

$K_{max} \leq \Delta(G_C) + 1$, where $\Delta(G_C)$: max degree in G_C . (*Vertex Color*)

Frequency Assignment: Upper Bound

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Frequency Assignment: Evaluation

Parameters:

- Number of nodes: $n : 200$, square region of area $A : 200 \times 200$
- Shortest Path Tree, Unit Disk Graph
- Density: n/A

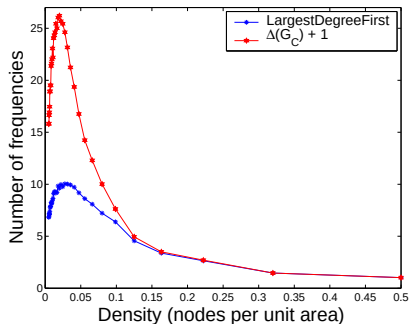


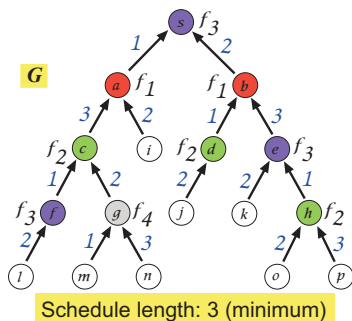
Figure: Comparison of the upper bound with the Largest Degree First scheme.

Time Slot Assignment

Time Slot Assignment

Algorithm: BFS-TIME SLOT ASSIGNMENT

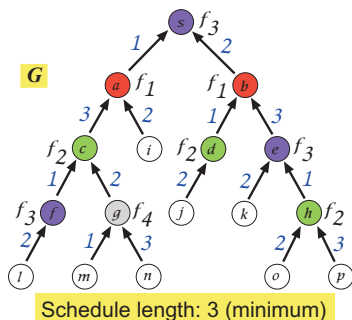
1. **while** $E_T \neq \phi$ **do**
2. $e \leftarrow$ next edge from E_T in BFS order
3. Assign minimum time slot to e respecting adjacency constraint
4. $E_T \leftarrow E_T \setminus \{e\}$
5. **end while**



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5. **end while**



Theorem

BFS-TIME SLOT ASSIGNMENT gives a schedule of minimum length $\Delta(T)$, where $\Delta(T)$: maximum node degree in the routing tree T .

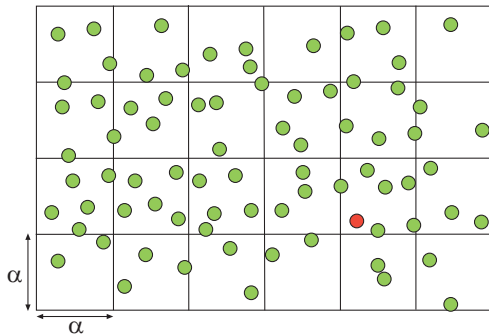
Proof: By induction.

Scenario Two:

Limited Number of Frequencies

Unit Disk Graph

Unit Disk Graph



Algorithm $\mathcal{G}$: Two Phases

- 1 Phase 1: Assign a frequency to each receiver in each cell.
- 2 Phase 2: Assign a time slot greedily to each edge.
e.g., schedule a *maximal number of edges* in each time slot.

Phase 1: Frequency Assignment

Goal:

Assign the frequencies to the receivers in such a way that the *maximum* number of nodes (*load*) transmitting on the *same frequency* is *minimized*.
(*Load-Balanced Frequency Assignment*)

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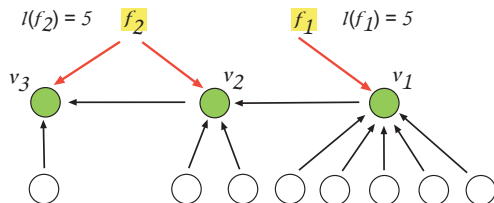
Load-Balanced Frequency Assignment is NP-hard.

Proof: Reduction from *Job Scheduling* problem [Garey, Johnson '79] on identical machines.

Phase 1: Frequency Assignment

Algorithm: FREQUENCYGREEDY

- 1 Sort the receivers in non-increasing order of in-degrees:
 $\text{deg}^{\text{in}}(v_1) \geq \text{deg}^{\text{in}}(v_2) \geq \dots \geq \text{deg}^{\text{in}}(v_n)$
- 2 Starting from v_1 , assign each successive receiver a frequency that has the *least load*, breaking ties arbitrarily.



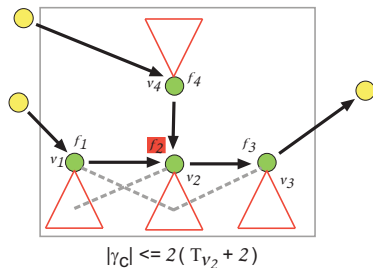
Phase 1: Frequency Assignment

Lemma:

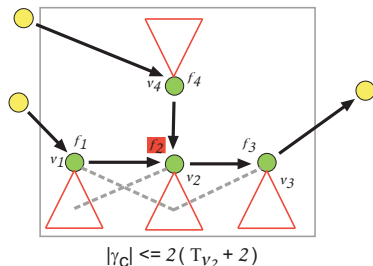
FREQUENCYGREEDY gives a $\left(\frac{4}{3} - \frac{1}{3K}\right)$ approximation on the min-max load, where K is the number of frequencies.

Proof: Follows from Graham's list scheduling according to *Longest Processing Time* first [Graham '69].

Phase 2: Time Slot Assignment



Phase 2: Time Slot Assignment

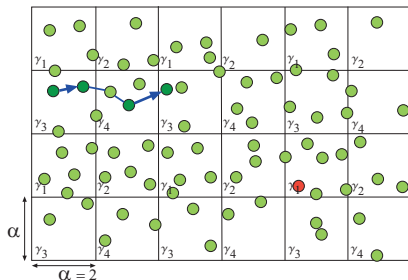


Lemma:

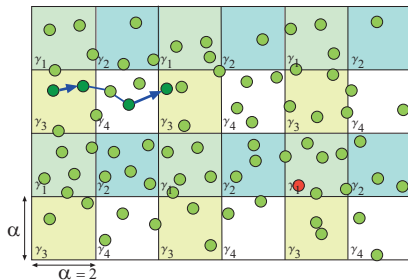
Suppose L_c^{FG} denote the load on the maximally loaded frequency in cell c achieved by **FREQUENCYGREEDY**.

Then, *any greedy* time slot assignment can schedule all the edges in c within number of time slots bounded by: $|\gamma_c| \leq 2 \cdot L_c^{FG}$, where γ_c denotes the *set of time slots* to schedule all the edges in c .

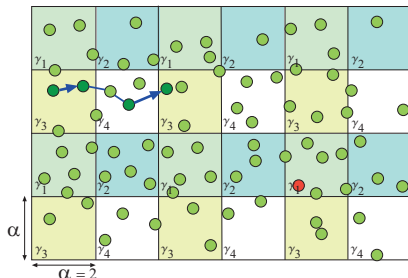
Phase 2: Time Slot Assignment



Phase 2: Time Slot Assignment



Phase 2: Time Slot Assignment



Lemma:

The *minimum schedule length* for the whole network is upper bounded by:

$$4 \cdot \max_c \{|\gamma_c|\}, \forall \alpha \geq 2$$

Proof: The colors represent sets of disjoint time slots. Follows from *Vertex Color* on rectangular grid.

Performance Bound

Algorithm $\mathcal{G}$

- 1 Run FREQUENCYGREEDY in each cell.
- 2 Run any greedy time slot assignment scheme.
e.g., schedule a *maximal number of edges* in each slot.

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Theorem

Algorithm $\mathcal{G}$ achieves a constant factor $8\mu_\alpha \cdot \left(\frac{4}{3} - \frac{1}{3K}\right)$ approximation on the optimal schedule length, where $\mu_\alpha > 0$ is a constant for a given cell size $\alpha \geq 2$ and for a given deployment of nodes.

(μ_α : maximum number of edges on the same frequency in any cell that can be scheduled on the same time slot)

Performance Bound

Proof Sketch:

Lower bound on *OPT*:

$$OPT \geq \frac{1}{\mu_\alpha} \max_c \{L_c^{LB}\}$$

Schedule length of *G*:

$$\begin{aligned}\Gamma_G &\leq 4 \cdot \max_c \{|\gamma_c|\} \\ &\leq 4 \cdot \max_c \{2 \cdot L_c^{FG}\} \\ &\leq 8 \cdot \max_c \left\{ \left(\frac{4}{3} - \frac{1}{3K} \right) \cdot L_c^{LB} \right\} \\ &\leq 8\mu_\alpha \left(\frac{4}{3} - \frac{1}{3K} \right) \cdot OPT\end{aligned}$$

Performance Evaluation

Parameters:

- Square region $A : 200 \times 200$, Shortest Path Tree, Unit Disk Graph

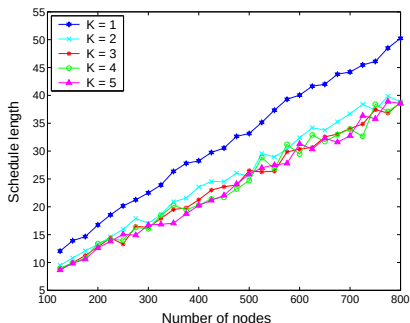


Figure: Schedule length of Algorithm $\mathcal{G}$ with network size for different number of frequencies K . Schedule length bottlenecked by the in-degree of the parents beyond a certain number of frequencies.

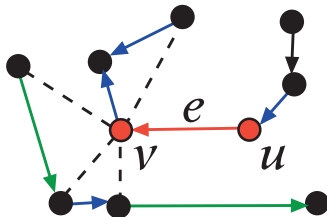
General Disk Graphs

General Disk Graphs

Notations:

- $r(u)$: Transmission range of node u
- $\ell(e)$: Length of edge e
- $I(e)$: Set of edges that are either adjacent to e or form a secondary interfering link with e
- $I_{\geq}(e)$: Subset of $I(e)$ with disks larger than e [SIGMETRICS'05]

$$I_{\geq}(e) = \{e' = (u', v') : e' \in I(e), \ell(e') \geq \ell(e)\}$$



ILP Formulation of Frequency Assignment

0-1 ILP:

- **Goal:** Minimize the maximum number of edges that interfere with any given edge.
- Define **indicator variables** X_{vk} for edge $e = (u, v)$ as:

$$X_{vk} = \begin{cases} 1, & \text{if } v \text{ is assigned frequency } f_k \\ 0, & \text{otherwise} \end{cases}$$

A frequency assignment is a **0-1 assignment** to $X_{vk}, \forall e, \forall f_k$.

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A frequency assignment is a **0-1 assignment** to $X_{vk}, \forall e, \forall f_k$.

- For every $e = (u, v)$ on f_k , define Z_{ek} as the total number of edges in $I_{\geq}(e)$ that are also on f_k

$$Z_{ek} = \sum_{e'=(u',v') \in I_{\geq}(e)} X_{v'k} = \sum_{v'} n(e, v') X_{v'k},$$

where $n(e, v') = |\{e' \in I_{\geq}(e) : \ell(e') \geq \ell(e)\}|$

ILP Formulation of Frequency Assignment

0-1 ILP:

Minimize λ

subject to :

$$\forall e = (u, v), \forall f_k : \sum_{v'} n(e, v') X_{vk} \leq \lambda \quad (1)$$

$$\forall e = (u, v) : \sum_{f_k} X_{vk} = 1, \quad (2)$$

$$\forall e = (u, v), \forall f_k : X_{vk} \in \{0, 1\} \quad (3)$$

- Solve the **Linear Relaxation** (LP) by modifying constraint (3)
- **Randomized Rounding**: Assign $Y_{vk} = 1$ with probability X_{vk}^* , where X_{vk}^* are the **optimal fractional** LP solutions and Y_{vk} are the new **integral** random variables.

ILP Formulation of Frequency Assignment

Lemma:

Let Y_{vk} be the **rounded** solutions, as described above. Then,

$$\max_{e, f_k} \left\{ \sum_{e'=(u', v') \in I_{\geq}(e)} Y_{v'k} \right\} = O(\Delta(T) \log n \cdot \lambda^*),$$

with probability at least $(1 - 1/n)$.

Proof. Apply **Chernoff bound**.

Time Slot Assignment

Lemma:

Let X_{vk} and Y_{vk} be defined as above. Then, $\Omega(\max_{e, f_k} \{Z_{ek}\})$ is a lower bound on the length of any schedule for the edges. Also, it is possible to schedule all the edges using $\max_{e, f_l} \{Z_{ek}\}$ time slots.

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Time Slot Assignment:

- Sort edges in **non-increasing** order of **lengths**: $\ell(e_1) \geq \ell(e_2) \dots$
- Assign the **smallest** available slot $t(e_i)$ to edge e_i such that:
 - $t(e_i) \neq t(e_j), \forall e_j$ with $j < i$ having the same receiver as e_i
 - $t(e_i) \neq t(e_j), \forall e_j$ that are assigned the same frequency as e_i

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- Assign the **smallest** available slot $t(e_i)$ to edge e_i such that:
 - $t(e_i) \neq t(e_j), \forall e_j$ with $j < i$ having the same receiver as e_i
 - $t(e_i) \neq t(e_j), \forall e_j$ that are assigned the same frequency as e_i

Theorem

The schedule constructed by time slot assignment strategy along with the frequency assignment using the above randomized rounding procedure results in a schedule of length $O(\Delta(T) \log n)$ times the optimum.

Conclusions and Future Directions

Conclusions:

- Multi-channel scheduling in sensor networks to eliminate interference
- Approximation algorithm on unit disk graphs
- Randomized algorithm on general disk graphs

Future Directions:

- Physical interference model
- What about minimizing the delay as well as maximizing the throughput? - **construct specific routing topologies and then do frequency and time scheduling**
- Not perfect aggregation - raw data collection